

Reasoning Reference: Number Series

Educational Series

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1 Introduction

A Number Series is a sequence of numbers following a specific logical pattern. The objective is to identify the underlying rule and find the missing number or the "wrong" number in the series.

2 Common Types of Series

2.1 Arithmetic Series (Difference-based)

The difference between consecutive terms is constant or follows a sub-pattern.

- **Example:** 2, 5, 8, 11, 14, ... (Pattern: +3)
- **Example:** 10, 15, 25, 40, 60, ... (Pattern: +5, +10, +15, +20)

2.2 Geometric Series (Ratio-based)

Each term is multiplied or divided by a constant factor.

- **Example:** 3, 6, 12, 24, 48, ... (Pattern: $\times 2$)
- **Example:** 1000, 200, 40, 8, ... (Pattern: $\div 5$)

2.3 Square and Cube Series

The numbers are directly squares/cubes or are related to them (e.g., $n^2 + 1$).

- **Example:** 1, 4, 9, 16, 25, ... ($1^2, 2^2, 3^2, \dots$)
- **Example:** 0, 7, 26, 63, ... ($n^3 - 1$)

2.4 Prime Number Series

The sequence consists of consecutive prime numbers.

- **Example:** 2, 3, 5, 7, 11, 13, 17, ...

2.5 Mixed/Combination Series

Uses more than one operation (e.g., multiply then add).

- **Example:** 3, 7, 15, 31, 63, ... (Pattern: $\times 2 + 1$)

3 The "Tier-Difference" Method

If a pattern is not immediately obvious, calculate the difference between terms, then the difference of those differences. This often reveals a hidden arithmetic progression.

4 Step-by-Step Strategy

1. **Check the Speed of Growth:**

- Slow increase/decrease $\rightarrow$ Addition/Subtraction.
- Rapid increase/decrease $\rightarrow$ Multiplication/Division.

2. **Identify Near-Squares/Cubes:** Check if terms are close to n^2 or n^3 .

3. **Look for Alternating Series:** Every second number might form a separate sub-series.

4. **Verify with the Final Term:** Once you find a pattern, apply it to the last given number to see if it matches the missing one.

5 Solved Examples

Example 1: Find the missing number: 4, 9, 20, 43, 90, ?

- $4 \times 2 + 1 = 9$
- $9 \times 2 + 2 = 20$
- $20 \times 2 + 3 = 43$
- $43 \times 2 + 4 = 90$
- $90 \times 2 + 5 = \mathbf{185}$

Example 2: Find the missing number: 1, 2, 6, 24, 120, ?

- $1 \times 2 = 2$
- $2 \times 3 = 6$
- $6 \times 4 = 24$
- $24 \times 5 = 120$
- $120 \times 6 = \mathbf{720}$